

Lunar ISRU Technology Demonstration

Team 05: Atmospheric Re-Entry Analysis and Primary TPS Design



Academic Supervisor: Dr. Paul Bruce
Dr. Koon-Yang Lee
Course: MEng Aeronautics with Spacecraft Engineering
Academic year: 2018/2019
Student: Guillermo Vargas
CID: 01193367
Group: LTD05
Date: 17/06/2019

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Abstract

A consistent methodology to conduct an analysis of the heat flux distribution over a general blunted axi-symmetric body of revolution has been developed to analyze and design the thermal protection system required for a Lunar sample-return Earth re-entry vehicle. The resulting peak temperatures and heat fluxes on the sphere-cone geometry devised require the use of charring ablative materials to dissipate the entering energy efficiently. Thus, a PICA heat shield was designed with a mass of 8.54 kg. Modelling of the ablative TPS phenomenology was then performed via a modified in-depth energy balance to model the ablative surface recession rate and in-depth temperature response, ultimately being employed as a validation tool for the sized TPS thickness by limiting the allowable bond-line temperature. Limitations and consequences of the approximate modelling and methodology employed is discussed throughout, with potential improvements being identified for further sizing optimisation iterations.

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Nomenclature

			V	Velocity	[m/s]
A	Reference Surface Area	[m ²]	β	Ballistic Coefficient	[kg/m ²]
C_D	Drag Coefficient	—	ϵ	Emissivity	—
C_P	Coefficient of Pressure	—	γ	Ratio of Specific Heats	—
C_p	Specific Heat Capacity	[J/kgK]	μ	Dynamic Viscosity	[Pa · s]
FPA	Flight Path Angle	[°]	ϕ	Local Inclination	[rad]
H	Specific Enthalpy	[J/kg]	ρ	Density	[kg/m ³]
H_g	Pyrolysis Enthalpy	[J/kg]	σ	Stefan-Boltzmann Constant	[W/m ² K ⁴]
$\bar{H}$	Partial Heat of Charring	[J/kg]	θ	Momentum Boundary Layer Thickness	[m]
k	Thermal Conductivity	[W/mK]	Subscripts		
M	Mach Number	—	0	Stagnation Conditions	
m	Entry Mass	[kg]	∞	Free-Stream Conditions	
$\dot{m}_g$	Pyrolysis Gas Mass Flow Rate	[kg/m ² s]	aw	Adiabatic Wall Conditions	
P	Pressure	[Pa]	$conv$	Convective	
Pr	Prandtl Number	—	e	Boundary-Layer Edge Conditions	
Q^*	Effective Heat of Ablation	[J/kg]	FR	Fay-Riddell Model	
$\dot{q}$	Heat Flux	[W/m ²]	hw	Hot Wall Conditions	
r	Recovery Factor	—	l	Laminar Conditions	
R_c	Radius of Curvature	[m]	max	Maximum Conditions	
Re	Reynolds Number	—	r	Radiative	
$\dot{S}$	Surface Recession Rate	[m/s]	T	Turbulent Conditions	
s	Wall Tangential Coordinate	[m]	w	Wall Conditions	
T	Temperature	[K]	Superscripts		
t	Time	[s]	*	Conditions at Reference Temperature	

1 Introduction

Manned missions to Mars are considered to be the next milestone for Space exploration, inspiring novel concepts such as NASA's plan for a lunar gateway for space exploration [1]. As proof of concept for many of the currently immature advanced technologies required for this purpose, DEM²ETER aims to design a sample-return Lunar mission to develop some of these technologies, including:

- In-situ-resource-utilization (ISRU) to extract and process the necessary propellant for the return from the Lunar surface back to Earth.
- Acquisition of material samples from the Lunar surface, and safe return to Earth of the re-entry capsule.

The latter of these objectives requires exhaustive analysis of the aerothermodynamic behaviour of re-entry bodies, so as to ensure the survivability of both the capsule and corresponding Lunar samples.

When entering any planetary atmosphere, a space vehicle must reduce its velocity by shaping the body to provide high aerodynamic drag, commonly adopting a blunted-cone shaped profile. As a result of both the blunted geometry and the environmental conditions, a detached bow shock appears ahead of the re-entry body, resulting in a temperature rise close to the vehicle's surface, as well as an interaction between the shock wave and boundary layer. Consequently, viscous dissipation within the boundary layer results in an increase in the wall temperature as heat is transferred through convection and radiation from the fluid. Subsequently, the heat is conducted from the surface of the body to the internal layers, hence the requirement for thermal protection systems (TPS) to protect the inner substructures from the extreme environmental conditions.

One of the greatest challenges when designing efficient aerospace TPS is determining the aerodynamic heating over the re-entry body during atmospheric flight, as well as the resulting internal thermal field which is generated.

The aim of this report is to present a comprehensive methodology for analysis of atmospheric re-entry performance, with an emphasis on dynamic pressure loads and aerothermodynamic heating on general, axisymmetric blunt bodies. In addition to this, thermal protection system design procedure is discussed, with a focus on ablative materials for high velocity Earth re-entry vehicles. A discussion of hypersonic heating

and TPS analysis methods is also presented. This report should be read along with [2] and [3] for a discussion on aerodynamic stability, CFD and launch and TDS strategies, as well as [4] considering non-atmospheric Lunar EDL procedures.

2 Re-Entry Environment and Trajectory Analysis

Behaviour of re-entry vehicles is strongly dependent on the particular atmospheric conditions considered. For Earth re-entry, the atmosphere is assumed to be stratified into five separate layers, with properties as defined in [5].

The first stage for atmospheric re-entry performance analysis requires a characterization of the entry trajectory. A characteristic parameter to describe such motion is the ballistic coefficient (β), defined by Equation 1 [6].

$$\beta = \frac{m}{C_D A} \quad (1)$$

Ballistic coefficients are effectively a measure of the effects of aerodynamic drag on the body's deceleration behaviour. Entry mass is defined by the requirements of the return mission, and $C_D = 1.05$ for the chosen 45° sphere-cone heat shield geometry, as described later in Section 5.2 [7]. The reference area is selected such that the vehicle can accommodate all necessary return instrumentation and payload, whilst minimizing size and mass.

The ballistic coefficient is often limited by the atmosphere on entry. A thinner atmosphere such as that in Mars requires low ballistic coefficients so that aerodynamic drag is capable of decelerating the body to within acceptable velocities for supersonic or subsonic TDS deployment and operation at an acceptable altitude, allowing enough time to prepare for landing without crashing [6]. This also influences whether a ballistic trajectory is allowed, or if a lifting trajectory must be employed. Lifting re-entry trajectories allow minimization of re-entry deceleration loads to almost any desired level, elongating the duration of re-entry flight to allow more time for smoother deceleration, but increasing complexity [8]. Although this is desirable in human return missions, the objective of DEM²ETER is to return lunar samples, hence higher g-loads are allowed. Therefore, a ballistic trajectory has been selected, reducing complexity and minimising entry flight time.

Following the methodology outlined in [9], time derivatives of the position and velocity vectors for the re-

entry body under a ballistic trajectory can be derived for a three-degree-of-freedom atmospheric flight model. Analysis of descent trajectories is dependent on vehicle motion relative to the particular atmosphere, hence requiring flight models based on an atmospheric fixed frame of reference. Since this rotates with the Earth, an Earth-centred, rotating frame of reference with spherical coordinates is employed to express the dynamic atmospheric flight equations. The dependency of this system of equations on β indicates the importance of this variable when characterizing re-entry bodies. Figure 1 presents the results of solving the dynamic equations for various ballistic coefficients, with the initial conditions given in Table 1 as per [10], where the altitude corresponds to the Kármán Line, since above this altitude aerodynamic drag and heating are negligible.

Table 1: Initial Atmospheric Entry Conditions.

Altitude [km]	Velocity [km/s]	FPA [°]
100	11	-6

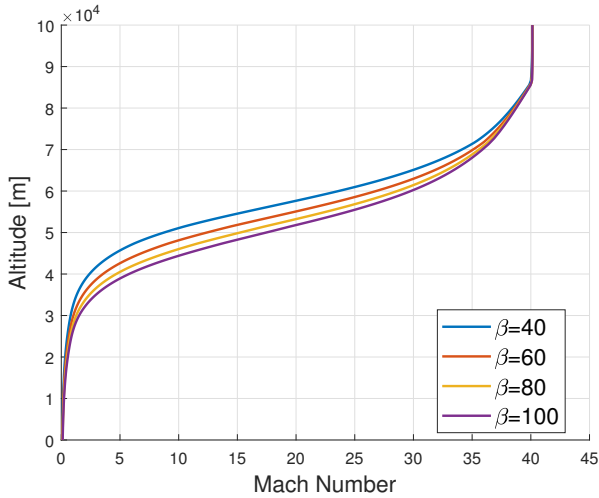


Figure 1: Re-Entry trajectories for varying β

2.1 Deceleration & Dynamic Pressure Loads

As can be observed in Figure 1, bodies with higher ballistic coefficients are less affected by atmospheric aerodynamic forces until lower in the atmosphere. Therefore, peak deceleration loads are expected to occur closer to the Earth's surface, as shown in Figure 2. Note that the vehicle is expected to sustain high g-loads on re-entry, of the order of 20g. Nonetheless, since the return capsule is non-manned and contains only lunar samples, these peak g-loads simply pose a structural constraint, limiting feasible TPS designs as discussed in Section 5.1.

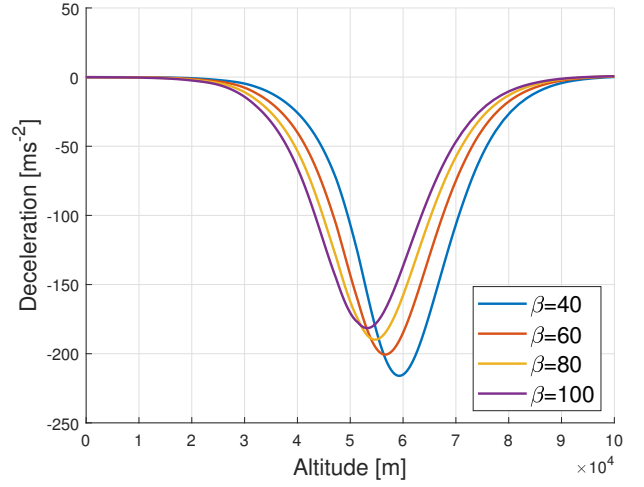


Figure 2: Deceleration Loads for varying β

Given peak deceleration loads occur lower down in the atmosphere for increasing β , it is known that bodies will be travelling at higher velocities closer to the Earth's surface, at which point air density will be greater. Therefore, peak dynamic pressure loads are expected to also occur lower down in the atmosphere, and have an increasing magnitude for larger ballistic coefficients, as observed in Figure 3. Overall, higher peak dynamic pressures are expected to result in more severe heating loads.

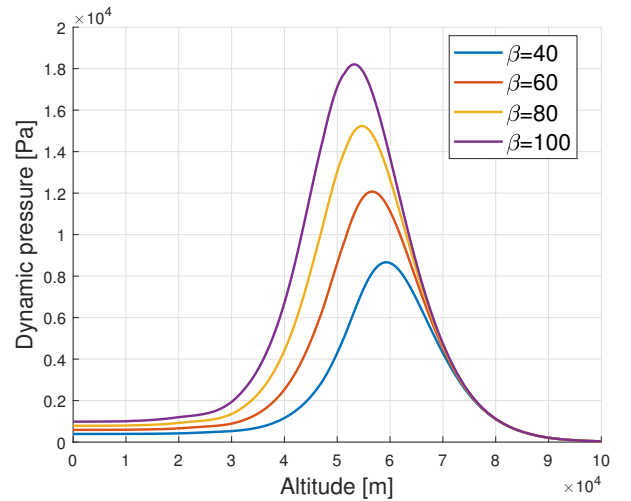


Figure 3: Dynamic Pressure Loads for varying β

Final Design Trajectory

The finalized capsule design had an entry mass of 34 kg [11], with a reference surface area of 0.5 m² (Section 5.2) and $C_D = 1.05$, resulting in a ballistic coefficient of $\beta = 64 \text{ kg m}^{-2}$. Figure 4 shows a representation of the final design trajectory for which heating analysis was performed. Peak dynamic pressure of $1.27 \times 10^{-4} \text{ Pa}$ is observed to occur at the same lo-

cation along the trajectory as the vehicle's maximum deceleration, with a magnitude of 198 ms^{-2} .

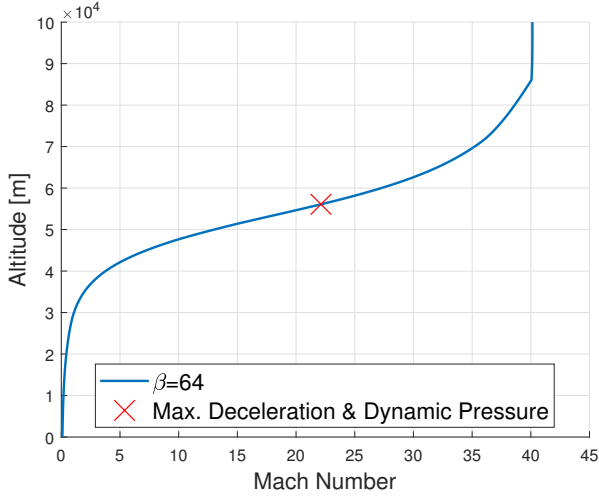


Figure 4: Final descent trajectory for re-entry vehicle.

3 Hypersonic Analysis

3.1 Modified Newtonian Theory

Once the final descent trajectory of the re-entry vehicle is known, the pressure distribution around the body must be obtained. To this end, local surface inclination methods must be employed, in particular modified Newtonian Theory due to the blunt-nosed body geometry. Since the majority of the re-entry flight environment is within the hypersonic regime, the generated bow shock is close to the body's surface, such that the underlying assumption of fluid particles losing all perpendicular momentum when 'colliding' with the surface is valid. Note however that Newtonian Theory is highly inaccurate for low velocity flight, since the influence of the surface is felt by the fluid far upstream [12].

The pressure distribution at any arbitrary location around the three-dimensional body being considered must now be obtained. To this end, the surface parametrization methodology defined in [13] is applied, such that the local angle distribution is obtained. Hence, following the theory assumption that $C_P = 0$ on the leeward surface, the pressure coefficient distribution over the unshadowed region of the vehicle is computed using Equations 2 and 3, whereby C_P is seen to be a function of M_∞ and γ as well as the local inclination.

$$C_P = C_{P_{max}} \sin^2 \phi \quad (2)$$

$$C_{P_{max}} = \frac{2}{\gamma M_\infty^2} \left\{ \left[\frac{(\gamma + 1)^2 M_\infty^2}{4\gamma M_\infty^2 - 2(\gamma - 1)} \right]^{\frac{\gamma}{\gamma-1}} \cdot \left[\frac{1 - \gamma + 2\gamma M_\infty^2}{\gamma + 1} \right] - 1 \right\} \quad (3)$$

4 Heating Analysis - Procedure

High fidelity methods are required to capture full body effects on the convective heating experienced by the re-entry vehicle. To this respect, a consistent procedure to conduct an analysis of the heat flux distribution over a general axi-symmetric body of revolution is now presented, where the particular heating model used at each location is observed to be dependent on both the local position with respect to the stagnation point and the particular flow regime. For this purpose, the Centerline Heating Model is adopted, whereby the body geometry along the windward centreline is treated as a known streamline along which aerodynamic heating is calculated, resulting in a simple, conservative, and low computationally complex scheme valid for 3D heat flux distribution over axi-symmetric geometries as the one being considered.

Initially, the centreline heating methodology requires computation of post-shock boundary layer edge conditions. Pressure, density, velocity, temperature, viscosity and Mach number at the boundary layer are therefore obtained from Equations 14-24 on [14], as described in [2].

4.1 Laminar & Turbulent Heat Flux

4.1.1 Transition

Before deriving separate models for laminar and turbulent convective heat flux, a flow transition criterion must be identified. Based on a study of hypersonic waveriders presented in [15] and discussed in [12], transition to turbulent flow can be considered to occur when the turbulent Reynolds number as defined in Equation 4 exceeds the local Reynolds number at any location over the body.

$$\log_{10}(Re_T) = 6.421 \exp(1.209 \cdot 10^{-4} M_e^{2.641}) \quad (4)$$

By identifying whether flow has transitioned at every position during the flight trajectory, the appropriate heat flux methodology can be calculated and stored.

4.1.2 Laminar Heat Flux

As outlined in [16], laminar heat transfer is modelled as a modified incompressible Blasius boundary layer relation, where compressibility effects are integrated into the model using Eckert's reference enthalpy approach as in Equation 5, from which the other reference conditions follow as in [2] and [14].

$$\frac{T^*}{T_e} = 0.45 + 0.55 \frac{T_w}{T_e} + 0.16 \sqrt{Pr} \frac{\gamma - 1}{2} M_e^2 \quad (5)$$

Therefore, convective heating calculations are computed by relating heat transfer to known skin-friction through a Reynolds analogy form via the boundary layer momentum thickness and Re_θ . Taking s as the local coordinate along the body streamline, the boundary layer momentum thickness is defined in terms of reference parameters as in Equation 6.

$$\theta_L(s) = 0.664 \frac{\left(\int_0^s \rho^* \mu^* V_e y^2 ds \right)^{0.5}}{\rho_e V_e y} \quad (6)$$

Finally, laminar convective heat flux is given by Equation 7.

$$\dot{q}_L = \frac{0.22}{Re_\theta} \frac{\rho^*}{\rho_e} \frac{\mu^*}{\mu_e} \frac{H_{aw} - H_w}{Pr^{0.6}} \rho_e V_e \quad (7)$$

Given the dependence of the laminar heat flux on Re_θ , Equation 7 tends to ∞ as the local coordinate, s , tends to 0. Thus, the model is singular at the stagnation point, hence the requirement for a separate heat flux model at, and around, the stagnation point, as described in Section 4.2.

4.1.3 Turbulent Heat Flux

Following a similar procedure to laminar flow, the heat transfer for turbulent flow is also computed by using a skin-friction coefficient based on Re_θ and a modified Reynolds analogy form. For ease of clarity, the relevant relations are not included in this report, but Equations 5-14 in [16] present a systematic approach to determine the turbulent heat flux. Note that turbulent boundary layers result in enhanced mixing and delay of separation, thus resulting in an increase in the skin friction experienced by the body. Due to the Reynolds analogy on which the presented heating models are built on, increased skin friction coefficients result in greater heat flux to the body.

4.2 Stagnation Point Heat Flux

As described previously, off-stagnation point convective heat flux models become singular near the stagnation point, thus a separate model is required in

this region. Methods such as that by Sutton-Graves allow determination of the heat-flux at the stagnation point of a spherical body, albeit in a simplified form [17]. Other methods such as the Fay-Riddell methodology [18] capture the full flow physical behaviour, and allow for heat flux determination at, and around, the stagnation point. A variant of the Fay-Riddell equation as in Equation 8 is suggested in [14], and is preferred due to its dependence on variables and flow conditions currently known, as well as its slightly broader applicability [2].

$$\dot{q}_{FR} = \frac{0.763}{Pr^{0.6}} (H_{aw} - H_w) \sqrt{\frac{\rho_e \mu_e}{R_c} \left(\frac{2(P_e - P_\infty)}{\rho_e} \right)^{0.5}} \quad (8)$$

These relations are defined for blunt bodies, with an inversely proportional dependence on the nose radius of curvature, such that spherical bodies with high R_c are preferred to minimise heat flux. Similarly, Equation 8 presents the direct dependency on the temperature gradient across the boundary layer, where the adiabatic wall enthalpy is obtained via a recovery factor, r , which is itself related to the Prandtl number, acting as the similarity parameter, and approximated as $Pr \approx 0.71$ for a laminar boundary layer in hypersonic flows [12]:

$$r = \frac{H_{aw} - H_e}{H_0 - H_e} \approx \sqrt{Pr} \approx 0.843$$

Overall, a mix of methods must be used to accurately predict the heating over all regions of the body. At and around the stagnation point, the Fay-Riddell heat flux model is used, which transitions into either the laminar or turbulent streamline heat flux methods according to the local flow regime. The Fay-Riddell stagnation heating model is averaged into the streamline methods, where the exact method transition location is defined by comparing the relative magnitude and location of applicability of the various models, so as to obtain a conservative overall heat flux distribution throughout the whole body profile.

4.3 Radiative Heat Flux

On return to Earth from lunar missions, vehicles are subjected to radiative heating from the incandescent shock-layer gases in the stagnation-point region [19]. Although the previous analysis has only considered convective heat flux transfer to the re-entry body, Tauber suggests that radiative heat flux must also be included at velocities greater than $9,000 \text{ m s}^{-1}$, where a heating rate expression as in Equation 9 is applied.

$$\dot{q}_r = CR_C^a \rho^{1.22} f(V) \quad (9)$$

For Earth entry, values of $f(V)$ are tabulated in [19], as well as suggesting values for the constants C and a . It is important to note that radiative heat flux does not rise linearly with increasing velocity, as shown by the tabulated values of $f(V)$. Thus, for lunar re-entry velocities the effect of radiative heat flux is minimal as compared to that of the previously computed convective heat flux, as can be observed in Figure 5, which graphically depicts this difference for the stagnation point as a representative location. Nonetheless, for the flight period at which body velocity is in excess of $9,000 \text{ m s}^{-1}$, radiative heat flux is added to the convective heat flux for completeness and accuracy. Note however that this is an important consideration for higher entry velocity missions, as could be a Mars return profile.

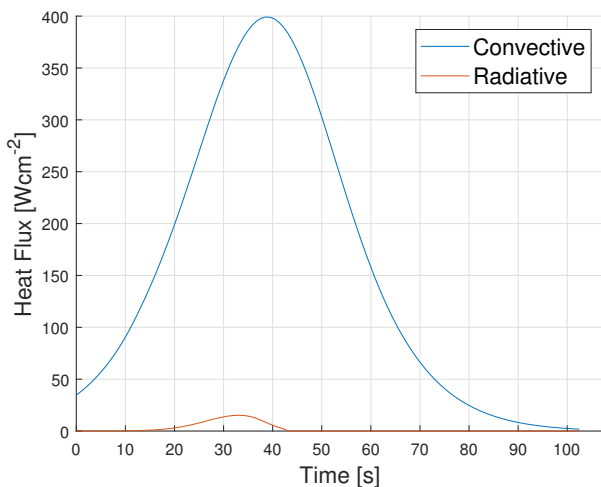


Figure 5: Comparison of stagnation point convective and radiative Heat Flux.

4.4 Wall Temperature Iteration

Throughout the heating analysis, all heat flux models considered present a direct dependence on the wall temperature, which is initially unknown. Therefore, the heating analysis over the body is necessarily an iterative method, whereby the wall temperature is initially assumed to be equal to the boundary layer edge temperature, and calculated iteratively until convergence between consecutive values of the calculated wall temperature is obtained.

The wall temperature varies throughout the entry trajectory, and peak wall temperatures are a key consideration when selecting the thermal protection system required. To this end, radiative equilibrium is employed to calculate the wall temperature distribution. This methodology assumes an energy balance whereby the convective heat transfer to the body is balanced by the radiative cooling from the surface,

obtained using the Stefan-Boltzmann law, such that $\dot{q}_{conv} = \epsilon\sigma T_w^4$. This results in an equilibrium temperature as in Equation 10.

$$T_w = \sqrt[4]{\frac{\dot{q}_{conv}}{\epsilon\sigma}} \quad (10)$$

4.5 Total Heat Load

Once the iterative procedure to obtain wall temperatures and convective heat flux to the re-entry body is finalised at discrete points along the descent trajectory, the resulting net heat flux along the windward streamline is integrated with respect to time over the entire trajectory to obtain the total heat load distribution. This concludes the heating analysis required for TPS design.

5 Re-Entry Vehicle Design

Before presenting the results obtained following the heating analysis procedure, this section aims to discuss the considerations behind the design of the re-entry vehicle geometry, with a brief review of novel technologies and preliminary profile determination.

5.1 Deployable Concepts

The requirement of achieving moderate ballistic coefficients for high payload mass missions to Mars has brought about the need to design novel EDL strategies. Deployable designs have appeared as alternatives to ‘conventional’ aero-shells, due to their reduced size and mass, resulting in lower operating costs [20]. Various concepts include mechanically deployable or inflatable heat shields, all of which aim to increase the entry surface area to limit the ballistic coefficient. Following recent research on the development of small re-entry spacecraft TPS designs, these concepts were initially considered due to the moderate ballistic coefficient of the mission.

Various concepts such as IRENE [21], PARES [22] or NASA’s ADEPT [23] present current attempts by various space agencies to develop large diameter, low β heat shields that limit aero-heating, allowing for fully reusable designs. Similarly, NASA’s HIAD [24] presents an inflatable alternative to increasing the vehicle diameter.

However, the maturity of these technologies is still moderate. Similarly, according to preliminary testing and validation of systems such as IRENE, application of deployable concepts is still limited to fairly low heat flux loads of approximately 30 W cm^{-2} [21]. Therefore, it was concluded that deployable TPS de-

signs, although promising, are not feasible for Earth re-entry vehicles from lunar missions, where the experienced heating over the body is high, causing structural and material limitations. Hence, a non-deployable, rigid heat shield design was adopted.

5.2 Heat Shield Geometry

Having selected a rigid heat shield, the profile must now be identified. As described in [25], objects with sharp corners in hypersonic flows cause Prandtl-Meyer expansion fans, leading to large velocity gradients near the body downstream of the corner. This therefore indicates regions with increased shear and friction coefficients, resulting in higher heat flux distributions by Reynolds analogy, as shown graphically in [26]. Consequently, sharp corners are to be avoided for bodies operating in hypersonic flow. Additionally, the Centreline Heating methodology employed for heating analysis does not capture the complex boundary layer effects generated, rendering these methods inaccurate for such geometry features.

Because of this, and inspired on classic re-entry body geometries such as the Soyuz and Shenzhou capsules, the re-entry body was firstly designed with a spherical profile to avoid increased sharp edge heating. Furthermore, the Fay-Riddell heating model provides an exact theoretical value for the heat flux at and around the stagnation point of a spherical profile vehicle, hence ensuring accurate predictions of heating. Although effective, this geometry results in excessive, unnecessary heat shield mass due to its large surface area. Furthermore, it does not necessarily provide optimal thermodynamic performance, as the study in [27] indicates. Therefore, a more efficient design was pursued with the objective of reducing overall mass for the required capsule volume, whilst preserving desired thermal performance.

To ensure the heating methods employed remain valid and accurate near the stagnation point, a sphere-cone profile was used, which is a common design for atmospheric re-entry vehicles [28]. In this way, the requirement of a spherical shape with an acceptable radius of curvature to minimise heat flux is maintained at the stagnation point, whilst minimising the overall heat shield mass and entry capsule volume dimensions. Several Mars missions such as Viking have relied on 70° sphere-cone capsules to land on the surface of the planet, where 70° is usually considered as the limit for allowable half-angle [29]. However, Mars has a thinner atmosphere, requiring Viking to employ lifting rather than ballistic trajectories. Therefore, a final 45° half-angle sphere-cone geometry was adopted, as a result of a trade-off among

hypersonic drag (heating), subsonic drag, and stability [30][31], as well as a compromise between minimising heat shield dimensions and mass whilst allowing enough volume for the in-capsule components [11]. The resulting design has an area of 0.5 m², with a base radius set to 0.4m, whilst the nose and shoulder radii were set to 0.2m and 0.02m as suggested by [29].

6 Heating Analysis - Results

Having defined the final geometry of the re-entry vehicle, the Centreline Heating analysis was now performed to obtain the corresponding thermal response throughout the entry trajectory, with particular emphasis on peak heating conditions. Following the heating analysis procedure, this section aims to present the results obtained and design the TPS accordingly, ensuring survivability of the entry vehicle under extreme thermal conditions. Note that the heating analysis methodology presented is only valid throughout the hypersonic flight regime ($M_\infty > 5$). However, from the trajectory analysis it was found that peak thermal conditions are obtained well within the hypersonic regime.

To this end, peak wall temperature, peak heat flux, and total heat load are defined as the limiting conditions for the return capsule, and are consequently considered to be the design parameters for which heat shield sizing is performed. Since peak thermal characteristics occur at the stagnation point of the vehicle, the presented results are those obtained at this location.

1. Peak Wall Temperature - Determines whether thermal insulating skins are sufficient, or if ablative skins are required for additional energy dissipation.
2. Peak Heat Flux - Determines which particular materials within the two main skin types are appropriate for the expected body heating.
3. Total Heat Load - The total heat to which the body is subjected throughout the whole flight phase determines the total heat shield thickness required to maintain a suitable inner capsule temperature.

Figure 6 presents the stagnation point heat flux and total heat load throughout the hypersonic flight regime of the re-entry vehicle. As can be observed, heat flux rises as the vehicle progresses through the increasingly dense lower atmosphere, reaching a peak heating value at an altitude of 64.8 km, where the velocity is 9.79 km s⁻¹. This is approximately 89% of the entry velocity, which is a typical value for a purely ballistic entry as given in [32]. Heat flux is then seen

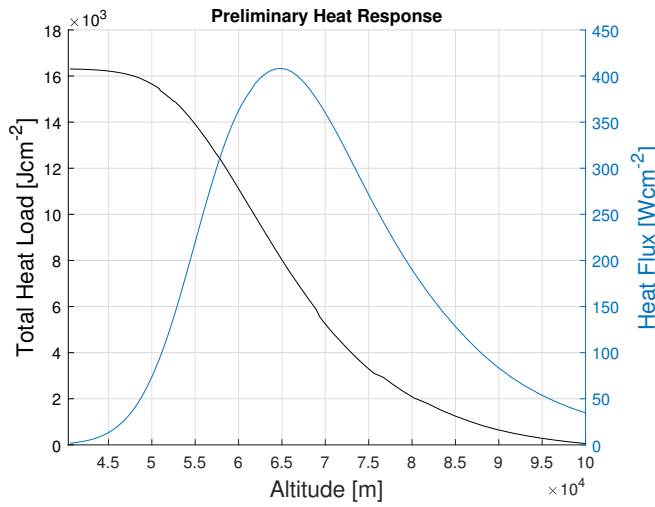


Figure 6: Stagnation Point Heat Flux and Total Heat Load throughout hypersonic flight regime.

to decrease, reaching negligible magnitudes as the vehicle approaches the supersonic regime. Similarly, the total heat load is seen to asymptote as the vehicle reaches the limit of the hypersonic regime. It is therefore expected that the vehicle will not be subject to considerable heating conditions at such low altitudes and velocities.

Additionally, the heat flux over the whole body must be known in order to optimally size a varying thickness heat shield, since off-stagnation heat loads are progressively lower the further away the point is from the stagnation point. Figure 7 presents a graphic representation of the thermal heat flux distribution over the whole re-entry body at the location of peak stagnation heat flux, indicating how the spherical nose section is subject to the most severe heating conditions, and so will require a thicker heat shield profile.

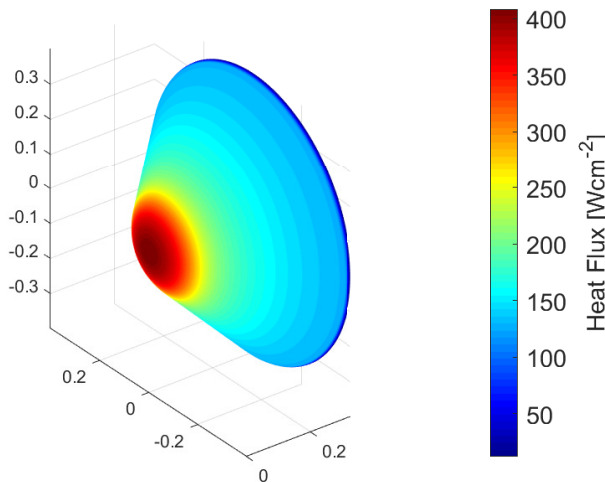


Figure 7: Heat Flux distribution over re-entry body at peak stagnation heat flux conditions.

Finally, stagnation point wall temperatures are presented in Figure 8, indicating a similar profile to that of heat flux distribution.

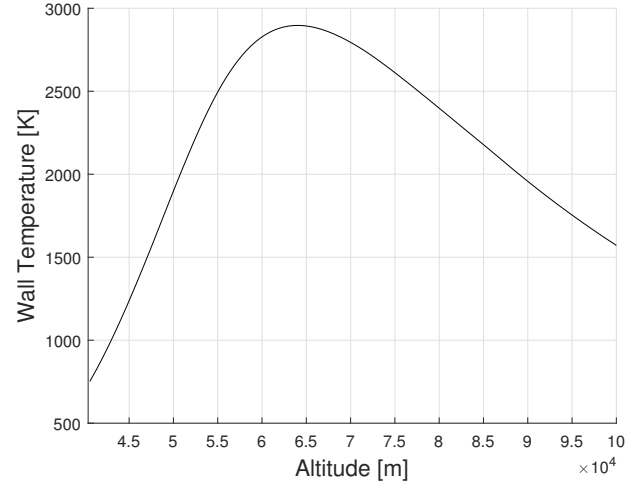


Figure 8: Wall temperature variation over hypersonic regime.

Table 2 summarizes the peak heating parameters obtained and later used in sizing the heat shield.

Table 2: Maximum Heating Conditions.

Peak Wall Temperature [K]	Peak Heat Flux [W/cm ²]	Total Heat Load [kJ/cm ²]
2,897	408	16.3

6.1 TPS Design and Sizing

TPS systems currently in use are mainly divided into reusable thermal heat sinks and ablative materials. The former of these is limited to low heat flux applications [33], whilst ablative TPS materials are capable of withstanding large heat fluxes and wall temperatures. From the results of the heating analysis, it was observed that the large heat load and wall temperatures obtained were above the operating environment of thermal skin materials. Therefore, it was concluded that the return capsule would necessitate an ablative heat shield design to dissipate all the incoming heat flux. Although thermal soak materials usually have lower thermal conductivity, the extreme temperatures and heating encountered along the descent trajectory indicate they would not withstand the high heat load experienced, so additional ablative energy dissipation mechanisms are required for the high entry velocity application being considered. Following these constraints, an estimate of the heat shield thickness for preliminary design is obtained following the empirical methods described in [34]. This

investigation presents mass estimating relationships to determine the required ablative TPS mass and thickness as functions of the total heat load and entry velocity. These relations are developed by predicting outputs from NASA’s FIAT program, achieved following multiple runs of FIAT with various initial entry conditions, within which the entry conditions of DEM²ETER return capsule lie. Historically, multiple classes of ablative materials have been successfully used as TPS materials for hypersonic bodies. This study considers three common, high performance ablative materials, namely Phenolic Impregnated Carbon Ablator (PICA), Carbon Phenolic Atop Advanced Carbon-Carbon 6 (CP/ACC), and Silicone Impregnated Reusable Ceramic Ablator (SIRCA), for which the main properties are summarized in Table 3. Additionally, Figure 9 shows a schematic of the resulting heat shield profiles obtained for the different ablative materials considered.

Table 3: Ablative Materials Properties.

	PICA	CP/ACC	SIRCA
Max. Allowable Heat Flux [W/cm ²]	1,200	30,000	100
Density [kg/m ³]	274	1,430	259
Min. Thickness [cm]	3.27	2.27	0.518

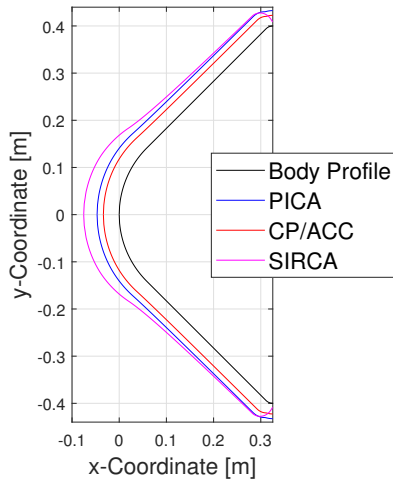


Figure 9: Heat shield profiles.

Analysis of Figure 9 allows determination of the total volume required by each heat shield design, and subsequently the various heat shield masses. Table 4 summarises the resulting mass for each material used.

Table 4: Ablative Material TPS Required Mass.

	PICA	CP/ACC	SIRCA
Mass [kg]	8.54	30.2	9.56

Firstly, it must be highlighted that SIRCA has only been used as an aft-body TPS material, and was found to be an unfeasible design material, since its maximum operating heat flux is below the peak heat flux experienced by the body on re-entry. Additionally, it is important to note that, although CP/ACC has the thinnest profile, its density is much higher than the other two, resulting in a greater overall heat shield mass. Therefore, with mass minimisation to reduce β and overall fuel requirements as the objective, a PICA profile was determined to be the optimal design for the considered re-entry vehicle due to its low density, good operating limit performance properties, and readily available database. The resulting maximum thickness of PICA at the stagnation point was found to be 46.8 mm.

As a concluding remark, it must be mentioned that these empirical methods are expected to slightly over-predict the required TPS. Nonetheless, many space missions over-design the heat shield due to the vital importance of the survivability of the re-entry vehicle. Similarly, it is a good preliminary design on which to perform through thickness performance analysis as presented in Section 7.3, after which the sizing can be refined at later iteration stages.

Further details on ablative phenomenology, as well as an explanation of the modelling of ablative materials with the objective of through-thickness TPS analysis will be further explained in Section 7, to serve as a validation tool for the presented preliminary empirical sizing.

7 Ablative TPS Performance Analysis

Following numerous theoretical, numerical and experimental methodologies for ablative modelling found in literature and based on common physical foundations, this section aims to describe the ablative TPS phenomenology, followed by a brief summary of the common method of analysis, and finally a presentation of a reduced complexity model as employed for the preliminary performance analysis of the empirically obtained design. Note that although the following methods are originally intended for TPS design and sizing, they will be used as a validation tool for the previously obtained heat shield profile. Further refining of these methods could be employed at a later stage to optimise TPS design.

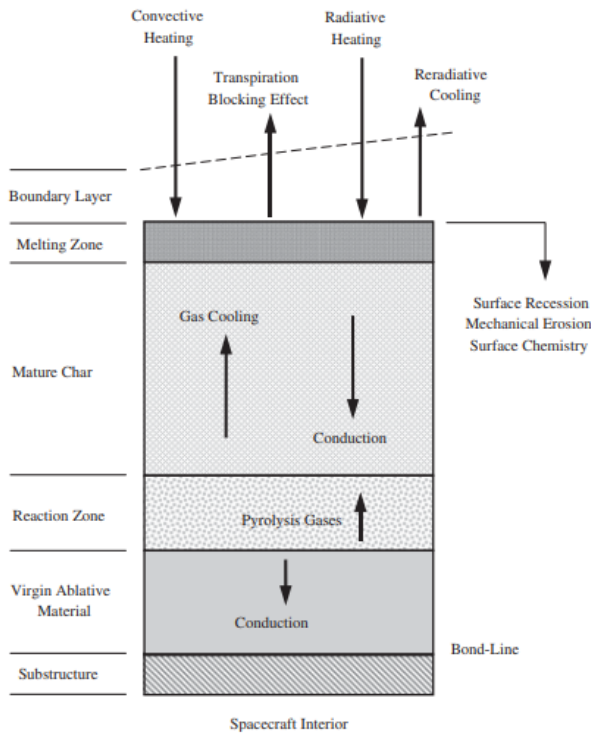


Figure 10: Schematic of main charring ablative phenomena.

7.1 Ablative TPS Phenomenology

The entire ablation process involves numerous physical and chemical phenomena whereby energy is dissipated by material changes of state. In general, ablative materials are classified as charring or non-charring as a defining characteristic, where charring materials such as PICA undergo chemical reactions and pyrolyse internally, whilst reactions are exclusively localized on the surface of the material for non-charring materials [35]. A schematic of the main physical and chemical phenomena undergone by charring ablative materials is shown in Figure 10.

Following the temperature rise near the body's surface due to bow shock and boundary layer interactions, heat is conducted through the internal layers of the charring TPS material. Above a threshold heating, the virgin material begins to change state via melting and sublimation phenomena in the presence of chemical reactions. This decomposes the charring material, generating gaseous products in what is known as pyrolysis reactions, and leaves a porous char residue with altered thermal properties. Eventually, pressure builds up in the reaction zone and pyrolysis gases are pushed through the porous char layer, where they further react, depositing residue in the porous char, and so further altering material thermal properties, whilst removing energy by convection. Pyrolysis gases finally reach the outer sur-

face and inject mass into the boundary layer, causing a blocking effect as the hotter boundary layer is displaced to reduce heat transfer to the heat shield. Mechanical erosion of the charred layer results in a surface recession, altering the thermal field within the material, as well as the effective body geometry.

Ablative charring materials are commonly used in atmospheric entry vehicles due to their ability to act as a thermal barrier, dissipating incoming thermal energy by chemical and physical processes, as well as absorbing a significant amount of heat by sacrificing a portion of the protecting material, whilst the generated pyrolysis gas flow creates a barrier effect, limiting the temperature rise of the vehicle substructures [35]. An accurate simulation model of the ablative phenomenology is therefore required, relying on accurate modelling of all the interacting processes and uncertainties [36].

7.2 Model and Governing Equations

Numerous models have been considered to develop an accurate ablation model, as described in [37] and [38]. Generally, the ablation process is modelled as a one-dimensional thermal energy transport in a multilayer stack of structures that can decompose in depth by ablating from the front surface. The TPS is modelled as a layer of ablative material bonded to the underlying substructure by a thin adhesive layer named the bond-line, which is considered to have a negligible thickness and so does not contribute to thermal insulation. A maximum allowable temperature for this layer is considered to be the design parameter, for which the minimum heat shield thickness is obtained. The ablative layer is itself generally subdivided into three distinct areas according to the running time conditions, namely the mature char, reaction zone and virgin material.

Early attempts to solve the thermo-chemical ablation phenomenon involved coupling a 1-D heat conduction equation with the heat of ablation, Q^* , to estimate surface recession. However, this is only valid for steady-state ablation. More rigorous approaches were subsequently introduced, adding complexity to the model by including in-depth decomposition of the material, as well as chemical processes and interactions. Subsequently, extensions to these works corrected the in-depth energy equation to account for pyrolysis gas generation and convection within the surface. An account of this work is detailed in [39], concluding with the common in-depth energy equation currently used as in Equation 11.

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} - \dot{q}_r \right) + (H_g - \bar{H}) \frac{\partial \rho}{\partial t} + \dot{S} \rho C_p \frac{\partial T}{\partial x} + \dot{m}_g \frac{\partial H_g}{\partial x} \quad (11)$$

From left to right, the energy balance equates the accumulation rate of thermal energy to the net rate of thermal energy that is transferred by conduction and internal radiation, the energy consumption during pyrolysis generation, the convective rate due to the mobile coordinate system, and the convective rate caused by the pyrolysis gases [36]. For ease of clarity, full model equations have not been included in this report. However, a full set of equations including modelling of the internal decomposition, internal mass balance, and varying thermal properties as a function of the local temperature can be found in [40] and [36].

These models have been converted to programs, of which two currently remain as the main analysis tools. Aerotherm's Charring Material Ablation (CMA) program developed by Moyer et al [41] is the older of these, whilst more recently NASA Ames developed the Fully Implicit Ablation and Thermal Response (FIAT) program. Although both rely on the same fundamental theory discussed, FIAT uses a fully implicit solution scheme, enhancing stability and convergence, as well as having a material database, automated mesh generation and thickness optimization [42].

7.3 Approximate Ablative TPS Analysis

An alternate methodology based on numerous assumptions is now presented and implemented. This allows preliminary estimation of the surface recession and through-thickness performance of the empirically designed ablative TPS system. Note that, although this formulation is based on early models of ablation where chemical decomposition and pyrolysis gas flow are ignored, and so is not representative of the full physical behaviour of ablation, it does provide a close representation of the phenomena, and a systematic validation estimate of preliminary TPS design performance. The main assumptions, and corresponding pitfalls/consequences of the system implemented are detailed throughout. Also, when considering ablative TPS thickness and recession rate, the ballistic coefficient and exact geometry of the body are assumed to remain constant throughout the descent trajectory, due to the comparably small mass loss and small geometrical surface recession.

Surface Recession Rate

Surface recession rates at each instant of time throughout the re-entry trajectory have been determined using Equation 12, whereby this formulation for ablation rate is only valid when considering steady-state ablation. However, steady-state ablation is difficult to achieve in the transient aerothermodynamic conditions of atmospheric entry [43]. As a result, Equation 12 leads to conservative results, generally over-predicting the actual amount of ablated material. Therefore, to account for this and minimize the error, a recession threshold temperature of 2,000 K for PICA has been incorporated, below which the recession rate is set to zero [44].

$$\dot{S} = \frac{\dot{q}_{hw}}{\rho Q^*} \quad (12)$$

This formulation employs the effective heat of ablation, Q^* , which is a function of the convective heat flux on the body, and a thermal property of PICA, valid only during steady-state ablation. Experimentally measured values for Q^* have been obtained from Figure 4 in [35]. Additionally, since the recessing material is from the charred layer, density has been set to the charred PICA density of 96 kg m^{-2} , as given in Table 5. Integrating the resulting recession rate profile during flight for $T_w > 2,000 \text{ K}$ as presented in Appendix A over time gives a recession distribution over the capsule forebody as shown in Figure 11, with a maximum total recession of 13.6 mm of the ablative surface at the stagnation point. This is consistent with similar ablative re-entry missions as Stardust, as seen by FIAT predictions in Fig. 3b in [42]. This thickness recession is later incorporated in the coordinate system modelling when determining the through-thickness temperature variation.

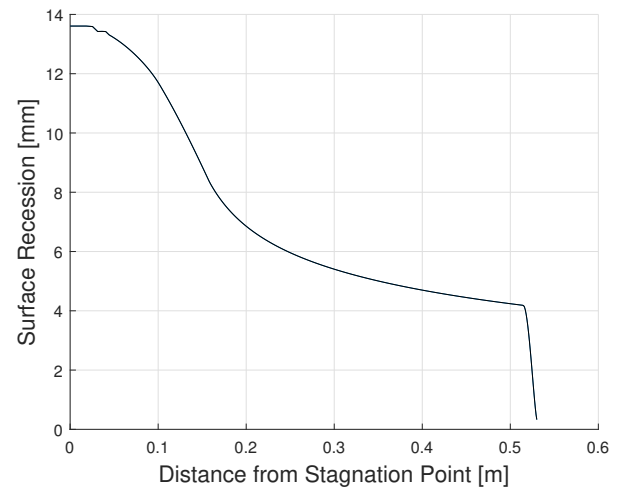


Figure 11: Surface recession distribution over TPS forebody.

In-Depth Temperature Response

The TPS comprises an ablative PICA material layer which is bonded to the capsule substructure through a thin adhesive layer at the bond-line, for which a maximum allowable temperature is defined by the material substructure. Therefore, the objective of determining the in depth-temperature behaviour is to ensure that the bond-line temperature remains below this specified limit. For the inner substructure material defined by the Structures Team [11], the maximum allowable temperature is given as 500 K.

An in-depth thermal response relation following a modified one-dimensional heat conduction model as given by Equation 13 has been adopted. This formulation has been used to analyse thermal flow entering the stagnation point of the re-entry vehicle, since this is the location where maximum heating occurs and hence is considered a dimensioning element [36].

$$\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{S} \rho C_p \frac{\partial T}{\partial x} \quad (13)$$

With respect to the full internal energy balance equation for an ablative, pyrolysing structure, Equation 13 ignores various forms of chemical fluxes entering the surface, as well as the terms corresponding to the energy consumption during pyrolysis gas generation, and the convective rate caused by the pyrolysis gases [40]. Note that the convective rate caused by the movement of the mobile coordinate system is included in the energy balance, having an effect on thermal behaviour when above the previously defined surface recession threshold temperature of 2,000 K. In addition to this, following determination of the surface recession, a local coordinate recession model has been implemented, whereby the surface recedes as time progresses, decreasing the TPS material thickness by the corresponding local surface recession.

The reduced thermal model is implemented by subdividing the ablative layer model into two different zones, namely a mature char and a virgin material zone, where it has therefore been assumed that both the reaction zone and bond-line thicknesses are negligible. Furthermore, PICA thermal properties (ρ , C_p and k) vary depending on the particular zone, according to the mean values suggested in [36] and summarized in Table 5.

Table 5: Mean PICA Thermal Properties.

	ρ [kg/m ³]	C_p [J/kg K]	k [W/m K]
Virgin	240	1800	0.30
Charred	96	4600	1.40

Implementation of this model employs a finite difference scheme with uniform nodal spacing, with assumption of a perfectly insulated inner substructure, resulting in a constant heat flux boundary condition at the bond-line ($\frac{dT}{dx} = 0$) [45]. Wall temperature and local surface recession rate are updated according to the heating analysis at every time-step, and the initial bond-line temperature is taken as 200 K, which is the atmospheric temperature at the Kármán Line. Note that this analysis is performed only during the hypersonic flight regime, taken as $M_\infty \geq 5$, since the heating analysis previously performed is no longer valid at lower velocities. However, heating at this point is low enough to avoid exceeding limiting temperatures.

From Table 5 it can be observed that the thermal diffusivity ($\alpha = \frac{k}{\rho C_p}$) of the charred layer is an order of magnitude greater than that of the virgin layer. However, the inner distribution of charred to virgin material varies throughout the trajectory. Given this uncertainty, it was assumed that these layers exist as a representative constant average ratio between each other, where the bond-line temperature increase is greatest for a fully charred heat shield. Nonetheless, it is not expected that the full thickness of the heat shield will char, given the over-design resulting from the empirical sizing used. Figure 12 presents a distribution of the bond-line temperature with time for various through-thickness compositions, where a limit condition of 20% virgin material remaining has been taken.

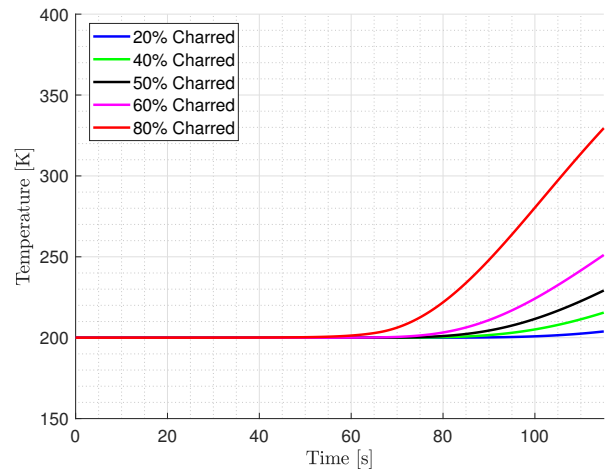


Figure 12: Bond line temperature distribution.

Figure 12 indicates how, even for a limiting condition where up to 80% of the TPS has charred, temperatures at the bond-line remain well below the maximum allowable temperature of 500 K for the inner substructure material. Further descent into the supersonic and subsonic regimes leads to negligible heating over the body, as compared to that encoun-

tered during hypersonic flight, therefore ensuring the survivability of the design. Note however that the margin with respect to limiting bond-line temperatures is still very large, highlighting the previously mentioned over-design of the empirical methods when sizing the thickness of the TPS.

As a concluding remark, a heat map of the through-thickness temperature variation obtained for the heat shield designed throughout the hypersonic flight regime of the re-entry trajectory is shown in Figure 13, for a representative average material composition of 50% virgin material remaining. This shows a graphic representation of the stagnation-point surface recession through time, as well as the temperature rise through the heat shield thickness, graphically indicating how the temperature rise is more pronounced within the charred layer, whilst only mild conduction ensues through the inner virgin layer.

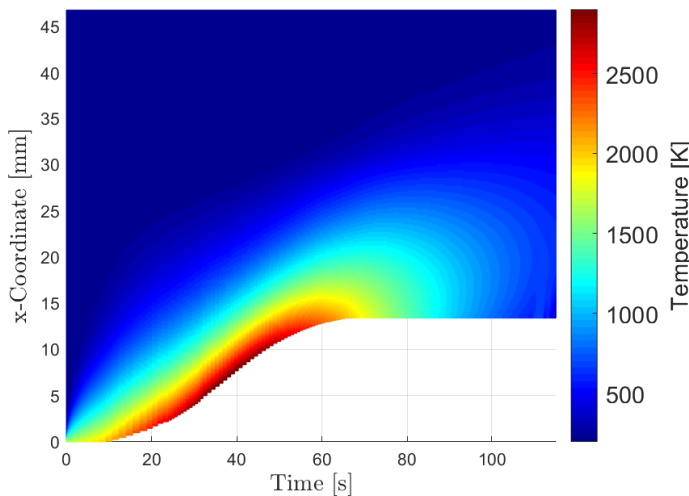


Figure 13: Stagnation point through-thickness temperature distribution for 50% virgin material composition.

8 Conclusions

Limitations and shortcomings of the presented methodology are discussed throughout, and further iterations with fewer assumptions would now be required to obtain a more accurate representation of the thermal behaviour of the TPS. In particular, adding the terms corresponding to energy consumption during pyrolysis gas generation to the in-depth energy equation would lead to even lower temperature rises, as energy would be employed in pyrolysis generation and convection rather than temperature rise. Nonetheless, the results obtained match initial expectations, and it was observed that the sizing methods resulted in TPS systems capable of withstanding the heating environment experienced on re-entry, albeit

over predicting the required TPS thickness.

In addition to a more comprehensive ablative modelling, the capsule geometry was taken as a 45° sphere-cone geometry throughout the heating trajectory. However, ablation of the surface causes changes in the profile and roughness of the surface, hence altering the overall heat distribution, requiring a dynamic modelling of the surface for a more accurate heating analysis. Additionally, further geometry optimization could be performed so as to alleviate the heat loads experienced as much as possible. Therefore an iterative procedure with alternative surface geometries as in [27] could result in an optimized profile with a milder overall heating environment.

Overall, when considering a Lunar sample return mission, re-entry into Earth's atmosphere poses a challenge on the design of a thermal protection system capable of protecting the vehicle from the high dynamic and thermal loads experienced on descent. To this end, a general methodology for heating estimation over a general axi-symmetric body of revolution with a blunted nose profile has been developed, with the objective of preliminary design of an effective TPS. Engineering methods employed in the Centreline Heating model and radiative equilibrium have been used to obtain a peak wall temperature of 2,897 K, indicating the requirement of ablative materials for additional energy dissipation. For a peak heat flux of 408 W cm^{-2} , PICA material was selected due to its low density, resulting in a heat shield mass of 8.54 kg via empirical methods based on FIAT. In order to validate the obtained result, an in depth analysis of the ablative methodology was performed, and a simplified ablative model was used to identify the TPS surface recession and through thickness wall temperature profile to ensure bond-line temperatures remain below substructure material limits. The analysis identified the design was sufficient for thermal protection. However, as predicted, it was found to be over-designed as bond-line temperatures remained approximately 150 K below the limiting material operating capabilities. Further investigation into the ablative model, and addition of dynamic rather than static behaviour of PICA thermal properties with temperature variations, as well as inclusion of additional chemical phenomena and pyrolysis gas effects would result in a more accurate model to use as a design methodology rather than a validation tool, allowing for optimisation of the TPS thickness profile as compared to the currently presented preliminary sizing. These could then be compared to FIAT results to assess the optimization accuracy of the devised model.

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Appendices

A TPS Surface Recession Rate

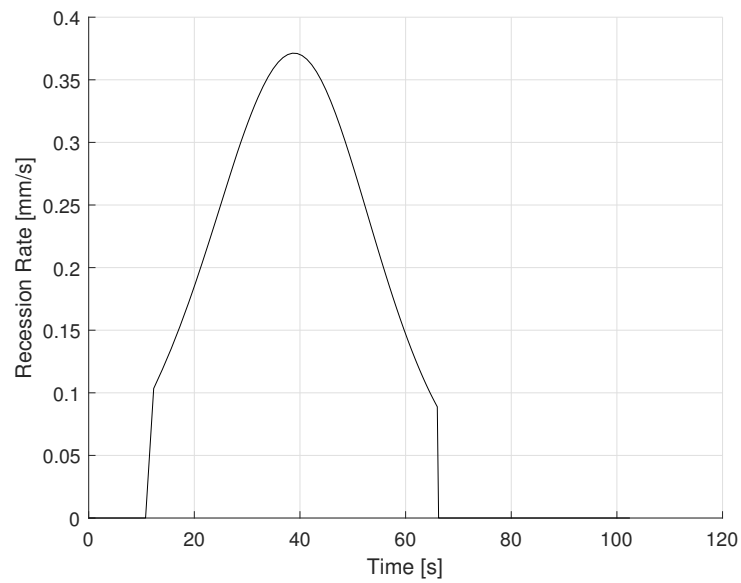


Figure 14: TPS surface recession rate over time.

B Re-Entry Capsule Surface Recession

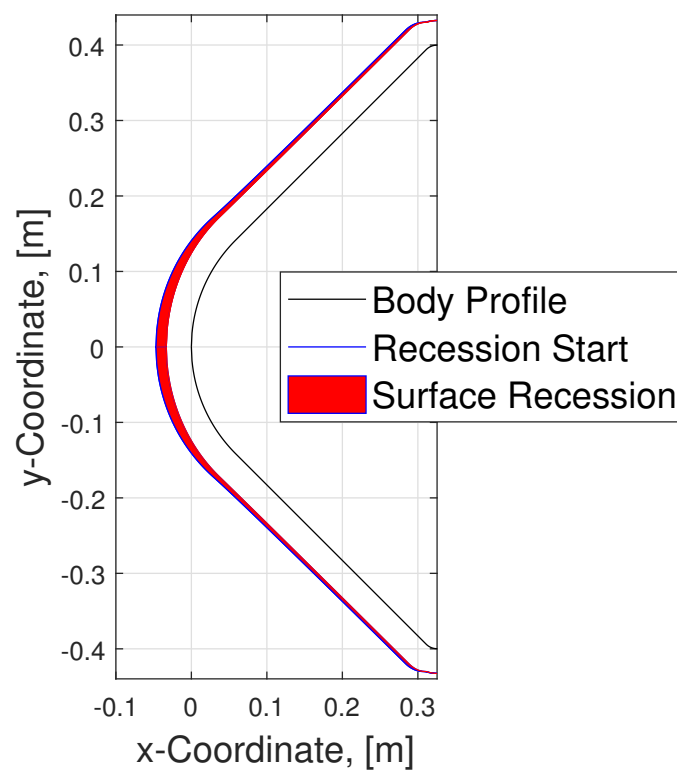


Figure 15: Graphical representation of the overall surface recession over TPS surface.